INTERNATIONAL GCSE MATHEMATICS A · MODULAR · UNIT 1 HIGHER

Top 20 Questions for Tomorrow

The questions that are coming

Twenty exam-style questions on the ideas that appear in every Unit 1 paper — histogram, probability, algebraic fractions and the rest. One question per page, full working space, full solutions at the back.



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20	86	100%	1
QUESTIONS	MARKS	ORIGINAL QUESTIONS	QUESTION PER PAGE

THE TWENTY

1 Histogram — draw and estimate
2 Histogram — read it back
3 Tree diagram without replacement
4 Three counters — exactly two the same
5 Probability with algebra
6 Algebraic fraction — simplify
7 Algebraic fractions — single fraction
8 Equation with algebraic fractions
9 Graph and a straight line
10 Bounds — suitable accuracy
11 Index equations
12 Surds
13 Completing the square
14 Cosine rule — angle and area
15 Sine rule — two steps
16 3D trigonometry — pyramid
17 Segment — area and perimeter
18 Perpendicular bisector
19 Recurring decimal
20 Venn diagram and probability

BOTTOM LINE Do each question in full, on its page, without looking. Then mark it with the solutions at the back (page 22). Every question you needed the solution for: do it again tonight.

1 Histogram — Draw and Estimate

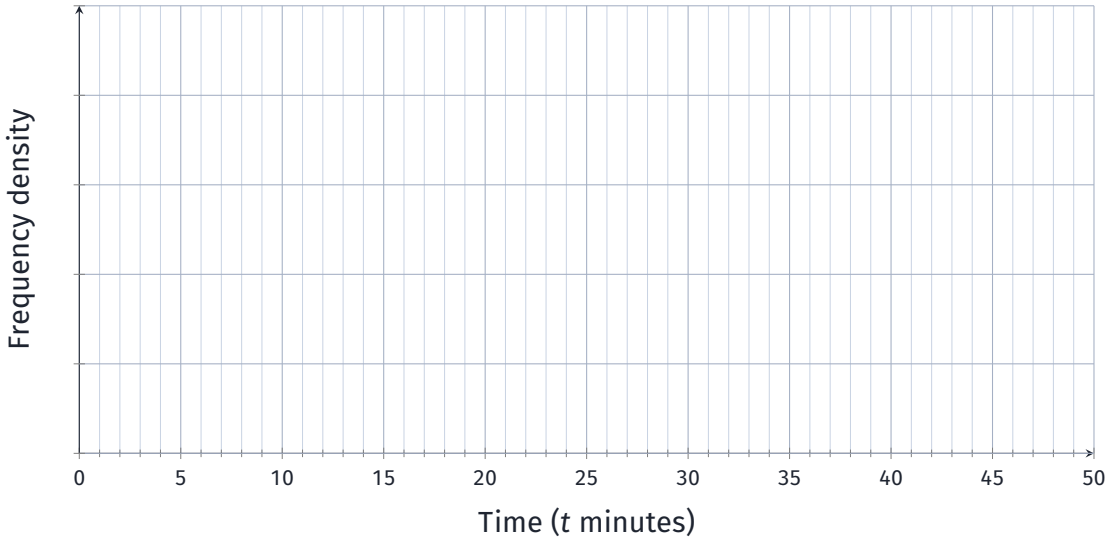
5 MARKS

Why it matters: a histogram has been in every Unit 1 paper from 2024 to 2026.

The table gives information about the times, t minutes, that 110 customers waited.

Time (t minutes)	Frequency
$0 < t \leq 5$	10
$5 < t \leq 15$	32
$15 < t \leq 20$	24
$20 < t \leq 30$	28
$30 < t \leq 50$	16

(a) On the grid, draw a histogram for this information. **(3)**



(b) Work out an estimate for the number of customers who waited between 12 and 25 minutes. **(2)**

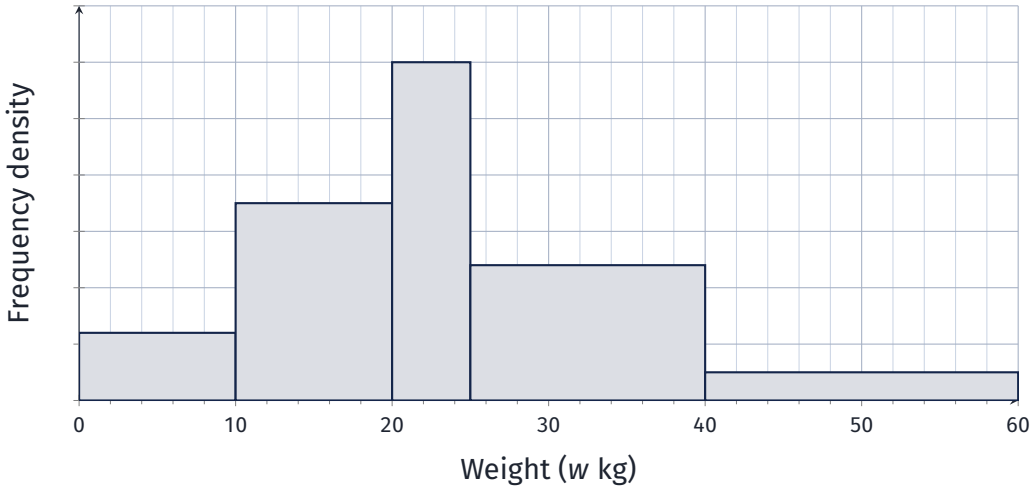
WORKING

2 Histogram — Read It Back

4 MARKS

Why it matters: "no scale on the axis" is the twist examiners love (May 2025, June 2026).

The histogram shows information about the weights, w kg, of some parcels.
12 parcels have a weight in the class $0 < w \leq 10$.



(a) Work out the total number of parcels. **(2)**

(b) Work out an estimate for the fraction of the parcels that weigh more than 30 kg. **(2)**

WORKING

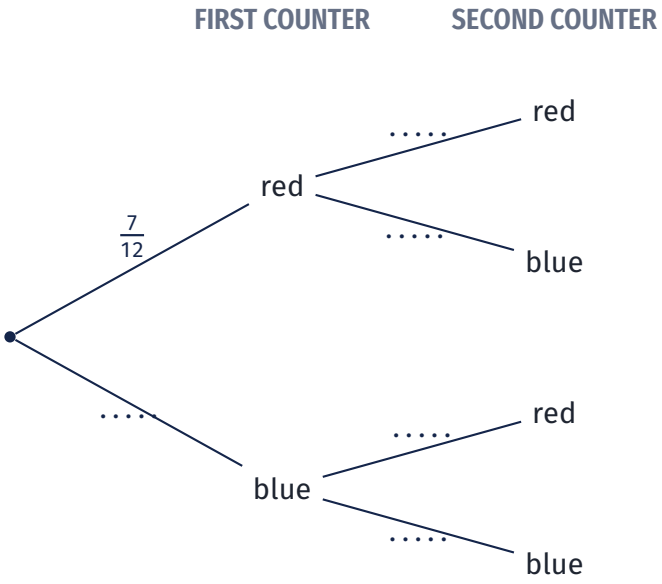
3 Tree Diagram — Without Replacement

4 MARKS

Why it matters: a tree diagram has been in every Unit 1 paper.

There are 7 red counters and 5 blue counters in a bag.
Lena takes at random a counter from the bag and does **not** put it back. She then takes at random a second counter.

(a) Complete the probability tree diagram. **(2)**



(b) Work out the probability that the two counters are different colours. **(2)**

WORKING

4 Three Counters — Exactly Two the Same

3 MARKS

Why it matters: the last-page probability question (May 2025 Q22, Nov 2025 Q23).

A box contains 9 counters: 4 are red, 3 are green and 2 are yellow.
Ravi takes at random three counters from the box, without replacement.
Work out the probability that exactly two of the three counters are the same colour. **(3)**

WORKING

5 Probability with Algebra

5 MARKS

Why it matters: June 2026 Q22 — a probability that turns into a quadratic.

A bag contains x red counters and 5 blue counters only. There are more red counters than blue counters.
Omar takes at random two counters from the bag, without replacement.
The probability that he takes one counter of each colour is $\frac{10}{21}$

(a) Show that $x^2 - 12x + 20 = 0$ **(3)**

(b) Find the number of red counters in the bag. **(2)**

WORKING

6 Algebraic Fraction — Simplify

3 MARKS

Why it matters: May 2025 Q19 and Nov 2025 Q28.

Simplify fully $\dfrac{2x^2 + 7x - 15}{4x^2 - 9}$ **(3)**

WORKING

7 Algebraic Fractions — Single Fraction

3 MARKS

Why it matters: "express as a single fraction" appears in most papers.

Express $\frac{5}{x+2} - \frac{3}{x-4}$ as a single fraction in its simplest form. **(3)**

WORKING

8 Equation with Algebraic Fractions

4 MARKS

Why it matters: fractions + quadratic formula in one question.

Solve $\frac{3}{x+1} + \frac{2}{x-2} = 1$

Give your solutions correct to 3 significant figures. Show clear algebraic working. **(4)**

WORKING

9 Graph and a Straight Line

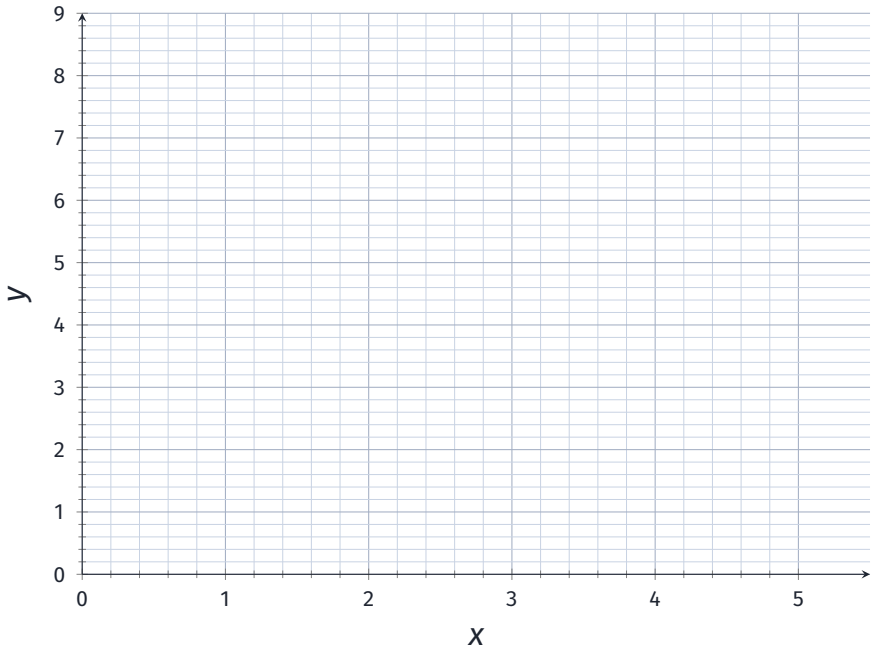
7 MARKS

Why it matters: every Unit 1 paper has a graph question.

(a) Complete the table of values for $y = x + \frac{4}{x}$ **(2)**

x	0.5	1	1.5	2	3	4	5
y	8.5		4.17		4.33		5.8

(b) On the grid, draw the graph of $y = x + \frac{4}{x}$ for $0.5 \le x \le 5$ **(2)**



(c) By drawing a suitable straight line on the grid, solve $\frac{4}{x} = 6 - 2x$ **(3)**

WORKING

10 Bounds — Suitable Accuracy

5 MARKS

Why it matters: June 2026 Q18 was exactly this (5 marks).

A metal block has mass 450 g, correct to the nearest 10 g.
The volume of the block is 62.4 cm^3, correct to 1 decimal place.

(a) Work out the upper bound for the density of the block. Give your answer to 3 s.f. **(3)**

(b) By considering bounds, write down the density to a suitable degree of accuracy. Give a reason for your answer. **(2)**

WORKING

11 Index Equations

5 MARKS

Why it matters: indices with an unknown power — every paper.

(a) Solve $27^{x} \div 9^{x-2} = 3^{10}$ **(2)**

(b) $\dfrac{\sqrt{2} \times 8^{x}}{16} = 2^{n}$ Express n in terms of x. **(3)**

WORKING

12 Surds

4 MARKS

Why it matters: a surd question has been in every Unit 1 paper.

(a) Expand and simplify $(2\sqrt{3} - 1)^2$ **(2)**

(b) Show that $\dfrac{12}{\sqrt{7} - 2}$ can be written in the form $a + b\sqrt{7}$ where a and b are integers. **(2)**

WORKING

13 Completing the Square

6 MARKS

Why it matters: the last question of 3 recent papers.

(a) Write $x^2 - 8x + 5$ in the form $(x - a)^2 - b$ where a and b are integers. **(2)**

(b) Hence solve $x^2 - 8x + 5 = 0$. Give your solutions in the form $p \pm \sqrt{q}$ **(2)**

(c) Express $3x^2 + 12x - 1$ in the form $a(x + b)^2 + c$ where a, b and c are integers. **(2)**

WORKING

14 Cosine Rule — Angle and Area

5 MARKS

Why it matters: non-right-angled triangles are in every paper.

The diagram shows triangle ABC.
$AB = 7$ cm, $AC = 9$ cm, $BC = 12$ cm.

(a) Work out the size of angle BAC. Give your answer correct to 1 decimal place. **(3)**

(b) Work out the area of the triangle. Give your answer correct to 3 significant figures. **(2)**

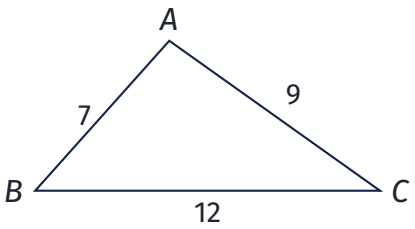


Diagram NOT accurately drawn

WORKING

15 Sine Rule — Two Steps

5 MARKS

Why it matters: find a side, then use it — the 5-mark trig question.

In triangle ABC, AC = 15 cm, angle BAC = 42° and angle ABC = 73°.

(a) Work out the length of BC. Give your answer correct to 3 significant figures. **(3)**

(b) Work out the area of triangle ABC. Give your answer correct to 3 significant figures. **(2)**

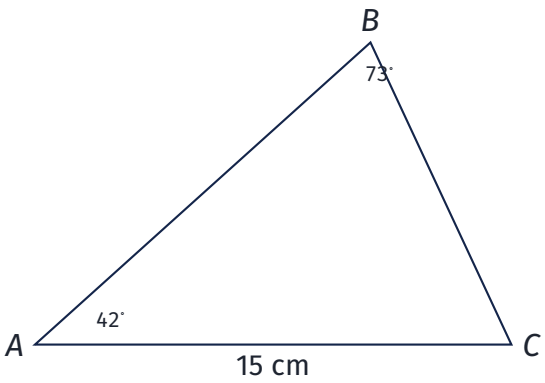


Diagram NOT accurately drawn

WORKING

16 3D Trigonometry — Pyramid

3 MARKS

Why it matters: 3D trig is in every paper and is often left blank — 3 easy marks.

VABCD is a pyramid with a square base *ABCD* of side 10 cm.
The vertex *V* is vertically above the centre *M* of the base, and *VM* = 12 cm.

Work out the size of the angle between the edge *VA* and the base *ABCD*.
Give your answer correct to 1 decimal place. **(3)**

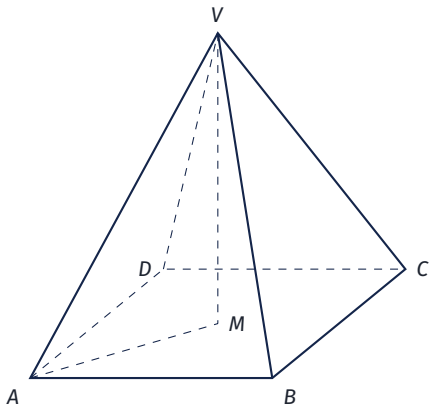


Diagram NOT accurately drawn

WORKING

17 Segment — Area and Perimeter

5 MARKS

Why it matters: sectors are in every paper; June 2026 Q23 asked for $a\pi + b$.

OAB is a sector of a circle, centre O, radius 10 cm. Angle $AOB = 60^\circ$.
The shaded region is the segment between the chord AB and the arc AB.

(a) Work out the area of the shaded segment. Give your answer to 3 s.f. **(3)**

(b) Find the perimeter of the shaded segment. Give your answer in the form $a + b\pi$. **(2)**

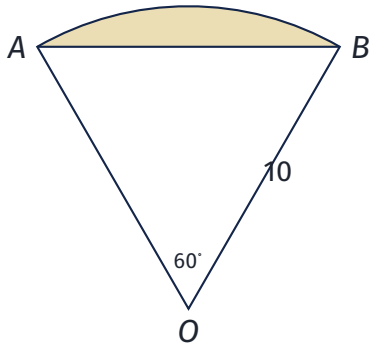


Diagram NOT accurately drawn

WORKING

18 Perpendicular Bisector

4 MARKS

Why it matters: May 2025 Q24 and Nov 2025 Q27 — the line through the other diagonal.

A is the point $(-2, 5)$ and B is the point $(6, 1)$.

Find an equation of the perpendicular bisector of AB.

Give your answer in the form $ax + by + c = 0$ where a, b and c are integers. **(4)**

WORKING

19 Recurring Decimal

2 MARKS

Why it matters: two quick marks (Nov 2025 Q17, May 2025 R).

Use algebra to show that $0.24\dot{5} = \frac{27}{110}$ **(2)**

WORKING

20 Venn Diagram and Probability

4 MARKS

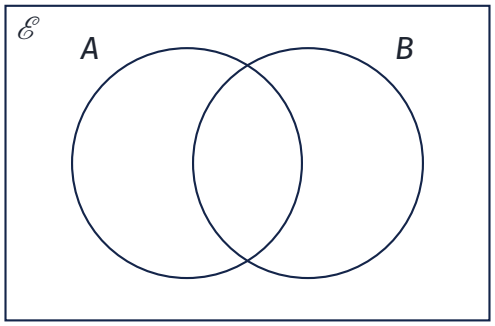
Why it matters: June 2026 Q5 — build the Venn diagram from clues.

$\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{1, 2, 3, 4, 6, 12\}$

$A \cap B = \{2, 4, 6, 12\}$

$B' = \{1, 3, 5, 7, 9, 11\}$



(a) Complete the Venn diagram. **(3)**

A number is chosen at random from $\mathscr{E}$.

(b) Find the probability that the number is in $A' \cap B$. **(1)**

WORKING

20 Full Solutions

Mark your work line by line. A method mark is for the right step even if the arithmetic slips — so check your steps, not only the answer.

1 Histogram — draw and estimate

1 **FREQUENCY DENSITIES**
$10 \div 5 = 2$, $32 \div 10 = 3.2$, $24 \div 5 = 4.8$, $28 \div 10 = 2.8$, $16 \div 20 = 0.8$. Bars of these heights, widths = the classes.

2 **12 TO 25 MINUTES**
$(15 - 12) \times 3.2 = 9.6$, 15–20: 24, $(25 - 20) \times 2.8 = 14$

ANSWER (b) $9.6 + 24 + 14 = 47.6 \approx 48$ customers

2 Histogram — read it back

3 **FIND THE SCALE**
The 0–10 bar is 12 parcels and has width 10, so its height is FD = 1.2: one big square = 1 unit.

4 **EACH BAR: FD × WIDTH**
$1.2 \times 10 = 12$, $3.5 \times 10 = 35$, $6 \times 5 = 30$, $2.4 \times 15 = 36$, $0.5 \times 20 = 10$

5 **MORE THAN 30 KG**
$(40 - 30) \times 2.4 = 24$ plus 10 = 34

ANSWER (a) 123 (b) $\frac{34}{123}$

3 Tree diagram — without replacement

6 **BRANCHES**
First: red $\frac{7}{12}$, blue $\frac{5}{12}$. After red: red $\frac{6}{11}$, blue $\frac{5}{11}$. After blue: red $\frac{7}{11}$, blue $\frac{4}{11}$.

7 **DIFFERENT COLOURS = RB + BR**
$\frac{7}{12} \times \frac{5}{11} + \frac{5}{12} \times \frac{7}{11} = \frac{35}{132} + \frac{35}{132} = \frac{70}{132}$

ANSWER (b) $\frac{35}{66}$

4 Three counters — exactly two the same

8 **ALL THREE DIFFERENT (R, G, Y IN ANY OF 6 ORDERS)**
$6 \times \frac{4}{9} \times \frac{3}{8} \times \frac{2}{7} = 6 \times \frac{24}{504} = \frac{144}{504} = \frac{24}{84}$

9 **ALL THREE THE SAME**
RRR: $\frac{4}{9} \times \frac{3}{8} \times \frac{2}{7} = \frac{24}{504}$; GGG: $\frac{3}{9} \times \frac{2}{8} \times \frac{1}{7} = \frac{6}{504}$; total $\frac{30}{504} = \frac{5}{84}$

10 **EXACTLY TWO THE SAME = 1 − (ALL DIFFERENT) − (ALL SAME)**
$1 - \frac{24}{84} - \frac{5}{84} = \frac{55}{84}$

ANSWER $\frac{55}{84}$

5 Probability with algebra

11 **ONE OF EACH COLOUR = RB + BR**
$\frac{x}{x+5} \times \frac{5}{x+4} \times 2 = \frac{10x}{(x+5)(x+4)}$

12 **SET EQUAL AND SIMPLIFY**
$\frac{10x}{(x+5)(x+4)} = \frac{10}{21} \Rightarrow 21x = x^2 + 9x + 20 \Rightarrow x^2 - 12x + 20 = 0$

13 **SOLVE**
$(x-2)(x-10) = 0$, $x = 2$ or $x = 10$. More red than blue (5), so $x = 10$.

ANSWER (b) 10 red counters

6 Algebraic fraction — simplify

14 **FACTORISE**
$2x^2 + 7x - 15 = (2x - 3)(x + 5)$; $4x^2 - 9 = (2x - 3)(2x + 3)$

ANSWER $\frac{x+5}{2x+3}$

7 Algebraic fractions — single fraction

15 **COMMON DENOMINATOR**
$\frac{5(x-4) - 3(x+2)}{(x+2)(x-4)} = \frac{5x - 20 - 3x - 6}{(x+2)(x-4)}$

ANSWER $\frac{2x - 26}{(x+2)(x-4)}$ (or $\frac{2(x-13)}{(x+2)(x-4)}$)

8 Equation with algebraic fractions

16 **MULTIPLY BY $(x+1)(x-2)$**

$3(x-2)+2(x+1)=(x+1)(x-2)$

17 **EXPAND AND COLLECT**

$5x-4=x^2-x-2 \Rightarrow x^2-6x+2=0$

18 **FORMULA**

$$x=\frac{6\pm\sqrt{36-8}}{2}=3\pm\sqrt{7}$$

ANSWER $x=5.65$ or $x=0.354$

9 Graph and a straight line

19 **TABLE**

$x=1$: 5; $x=2$: 4; $x=4$: 5. Plot all 7 points, join with a smooth curve (lowest point (2, 4)).

20 **MAKE THE LEFT SIDE THE GRAPH**

$\frac{4}{x}=6-2x$. Add x to both sides: $x+\frac{4}{x}=6-x$

21 **DRAW THE LINE $y=6-x$**

Through (0, 6) and (5, 1). It meets the curve at $x=1$ and $x=2$.

ANSWER (a) 5, 4, 5 (c) $x=1$, $x=2$

10 Bounds — suitable accuracy

22 **BOUNDS**

$445\le m<455$; $62.35\le V<62.45$

23 **UB DENSITY = UB MASS ÷ LB VOLUME**

$455\div 62.35=7.2975\ldots$

24 **LB DENSITY = LB MASS ÷ UB VOLUME**

$445\div 62.45=7.1257\ldots$

ANSWER (a) 7.30 g/cm^3 (b) 7 g/cm^3 — both bounds round to 7 to 1 s.f.

11 Index equations

25 **(A) BASE 3**

$3^{3x}\div 3^{2x-4}=3^{x+4}$, so $x+4=10$

26 **(B) BASE 2**

$\sqrt{2}=2^{1/2}$, $8^x=2^{3x}$, $16=2^4$: $2^{\frac{1}{2}+3x-4}$

ANSWER (a) $x=6$ (b) $n=3x-\frac{7}{2}$

12 Surds

27 **(A)** $(2\sqrt{3}-1)(2\sqrt{3}-1)$

$12 - 2\sqrt{3} - 2\sqrt{3} + 1 = 13 - 4\sqrt{3}$

28 **(B) CONJUGATE**

$$\frac{12(\sqrt{7}+2)}{(\sqrt{7}-2)(\sqrt{7}+2)} = \frac{12(\sqrt{7}+2)}{7-4} = 4(\sqrt{7}+2)$$

ANSWER (a) $13 - 4\sqrt{3}$ (b) $8 + 4\sqrt{7}$

13 Completing the square

29 **(A)**

$x^2 - 8x + 5 = (x-4)^2 - 16 + 5 = (x-4)^2 - 11$

30 **(B)**

$(x-4)^2 = 11 \Rightarrow x - 4 = \pm\sqrt{11}$

31 **(C)**

$3(x^2 + 4x) - 1 = 3\left[(x+2)^2 - 4\right] - 1 = 3(x+2)^2 - 13$

ANSWER (a) $(x-4)^2 - 11$ (b) $4 \pm \sqrt{11}$ (c) $3(x+2)^2 - 13$

14 Cosine rule — angle and area

32 **ANGLE A IS OPPOSITE THE 12**

$$\cos A = \frac{7^2 + 9^2 - 12^2}{2(7)(9)} = \frac{-14}{126}$$

33 **INVERSE COS**

$A = 96.379\ldots^\circ$ (obtuse — cos is negative)

34 **AREA**

$\frac{1}{2}(7)(9)\sin 96.379^\circ = 31.30\ldots$

ANSWER (a) 96.4° (b) $31.3\ \text{cm}^2$

15 Sine rule — two steps

35 **BC IS OPPOSITE 42°, AC IS OPPOSITE 73°**

$$\frac{BC}{\sin 42^\circ} = \frac{15}{\sin 73^\circ} \Rightarrow BC = \frac{15 \sin 42^\circ}{\sin 73^\circ} = 10.495\ldots$$

36 **THIRD ANGLE**

$C = 180 - 42 - 73 = 65^\circ$

37 **AREA**

$\frac{1}{2}(15)(10.495\ldots)\sin 65^\circ = 71.34\ldots$

ANSWER (a) 10.5 cm (b) 71.3 cm^2

16 3D trigonometry — pyramid

38 **HALF THE DIAGONAL**

$AC = \sqrt{10^2 + 10^2} = 14.142\ldots$, so $AM = 7.071\ldots$

39 **TRIANGLE AMV, RIGHT ANGLE AT M**

$$\tan\theta = \frac{12}{7.071\ldots}$$

ANSWER 59.5°

17 Segment — area and perimeter

40 **SECTOR**

$\frac{60}{360} \times \pi \times 10^2 = 52.359\ldots$

41 **TRIANGLE OAB**

$\frac{1}{2}(10)(10)\sin 60^\circ = 43.301\ldots$; segment $= 52.359\ldots - 43.301\ldots$

42 **PERIMETER**

Triangle OAB is equilateral, so $AB = 10$. Arc $= \frac{60}{360} \times 2\pi(10) = \frac{10\pi}{3}$

ANSWER (a) 9.06 cm^2 (b) $10 + \frac{10}{3}\pi$ cm

18 Perpendicular bisector

43 **MIDPOINT AND GRADIENT**

Midpoint $(2, 3)$; gradient $AB = \dfrac{1 - 5}{6 - (-2)} = -\dfrac{1}{2}$

44 **PERPENDICULAR GRADIENT**

2: $y - 3 = 2(x - 2) \Rightarrow y = 2x - 1$

ANSWER $2x - y - 1 = 0$

19 Recurring decimal

45 **LINE UP THE TAILS**

$x = 0.24545\ldots$, $10x = 2.4545\ldots$, $1000x = 245.4545\ldots$

46 **SUBTRACT**

$990x = 243 \Rightarrow x = \dfrac{243}{990} = \dfrac{27}{110}$

20 Venn diagram and probability

47 B **FROM** B'

$B = \{2, 4, 6, 8, 10, 12\}$

48 **REGIONS**

$A \cap B$: 2, 4, 6, 12; A only: 1, 3; B only: 8, 10; outside: 5, 7, 9, 11

ANSWER (b) $A' \cap B = \{8, 10\}$, probability $\dfrac{2}{12} = \dfrac{1}{6}$